

Procedure:

Adjust RV101 fully counter-clockwise. Apply ± 6 V to J102.
Do not connect AWG.

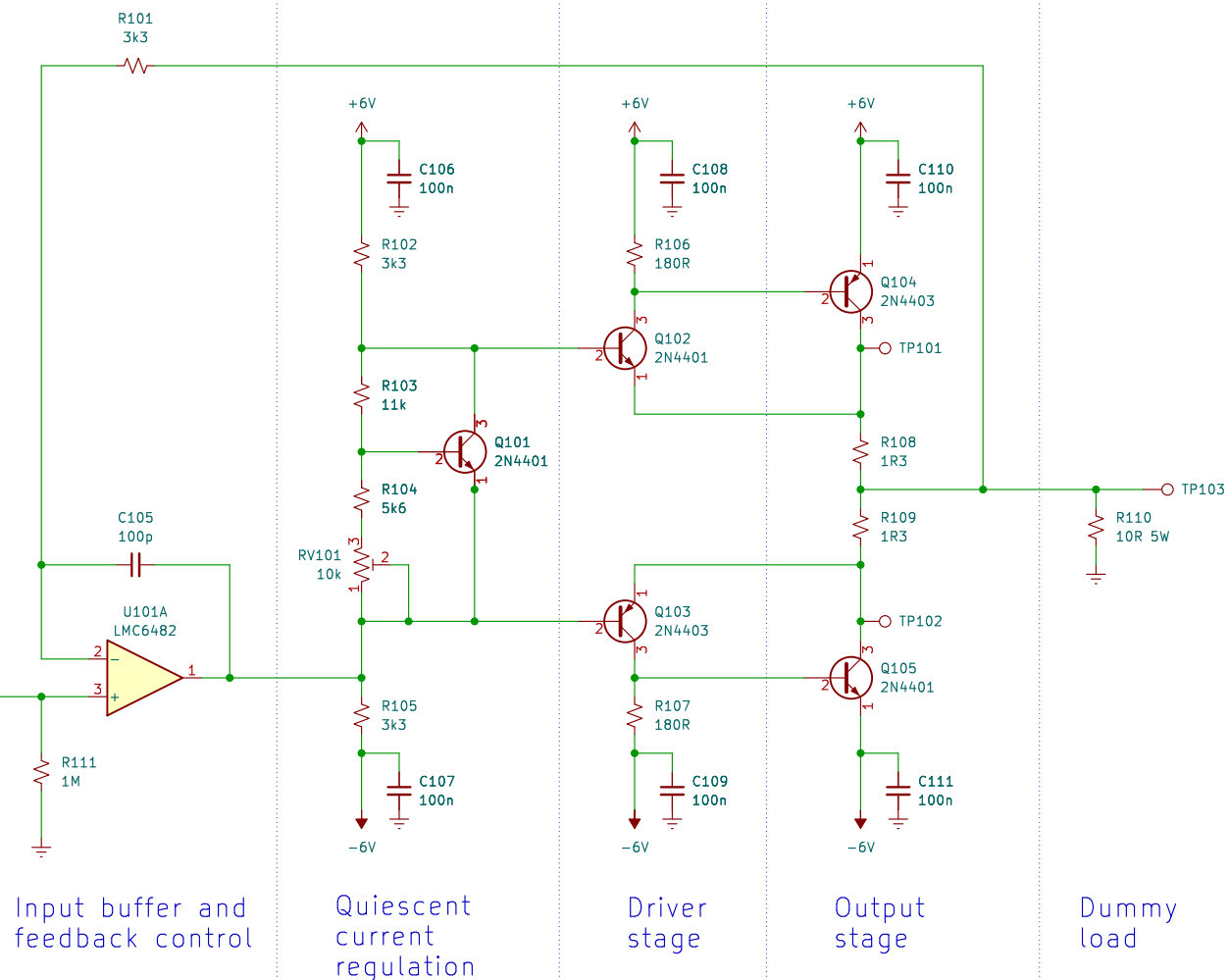
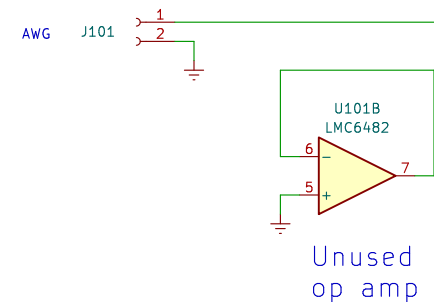
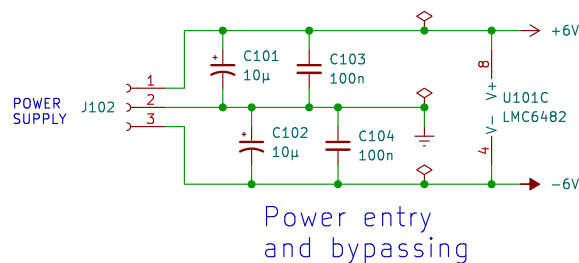
With millivoltmeter monitoring TP101–TP102, adjust for 65 mV.
(25 mA quiescent current.)

Connect AWG to J101 and oscilloscope to TP103.

Set AWG for 500 mV, 1kHz sine wave. Observe crossover distortion
(or lack thereof).

Increase AWG output level while observing on oscilloscope and
thermal camera. Quit when there is evidence of circuit clipping
or of overheating.

Try various frequencies from 20 Hz to 20 kHz – observe whether
circuit is fast enough.



Rev. B: [B5] Change R109/R109 to 1R3 from 1R2 because no 1R2 were on hand.

[A2] Change test procedure to reflect different R109/R109 values.

[C2] Add R111 to the input to ensure correct operation when AWG is disconnected.

Rev. C: [A1/A2] Change quiescent current in demo procedure to 25 mA
(and quiescent voltage to 65 mV) to match video narration.

C101, C102: Aluminum electrolytic, 15V or higher.
C103–C111: Ceramic, X7R or Y5V, 15V or higher.
R101–R109, R111: 1.4 W, 1% metal film.
R110: 10R, 5 W ceramic-encased wirewound.
RV101: 10k multiturn trimpot.



Transistors 101
Episode 20 – More Power
Kludges from Kevin's Cave

Sheet: /
File: SziklaiTestAmp.kicad_sch

Title: Demo amplifier – Sziklai push-pull outputs

Size: A Date: 2026-06-30
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Rev: C
Id: 1/1

Note: Do not try to run this amplifier with an actual speaker unless you're willing to sacrifice some cheap transistors. It is pushing the small signal-transistors right to their limit with a purely resistive load, and is likely to fail with a reactive load.